

EENG250 ELECTRIC CIRCUITS I · CHAPTER 1

Circuit Variables

Four quantities and one sign rule you will use in every chapter: charge, current, voltage and power.

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PREDICT FIRST

Your home has a 5 A generator subscription. Can you run a 2000 W kettle?

5 A × 220 V ≈ 1.1 kW. No: the breaker trips. The ampere size caps your **power** at any moment; the kWh meter counts your **energy** over the month.

TODAY

By the end of this session you can

1 • Units

Use SI units and the prefixes p, n, μ , m, k, M, G.

2 • Current and voltage

$$i = \frac{dq}{dt}, \quad v = \frac{dw}{dq}$$

3 • Power and energy

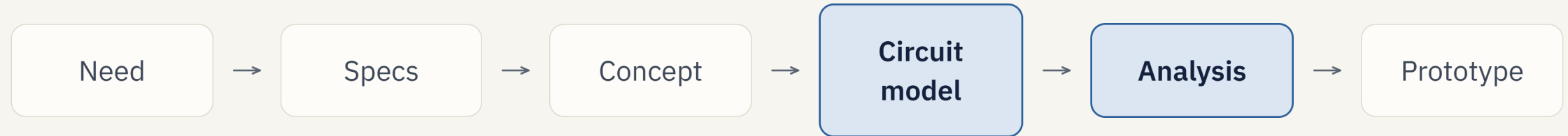
$$p = vi, \quad w = \int p dt$$

4 • The sign rule

Say from a diagram whether an element absorbs or delivers power.

CIRCUITS ARE MODELS

A circuit is a mathematical model of a real system



Measurements on the prototype send you back to refine the model. **This course is the two blue boxes:** we predict a circuit's behaviour from its voltages and currents before anyone builds it.

WHEN CIRCUIT THEORY APPLIES

The system must be small compared with the wavelength

$$\lambda = \frac{c}{f}, \quad c = 3 \times 10^8 \text{ m/s}$$

Nilsson's one-tenth rule: the system is **lumped** if it is smaller than one tenth of λ .

Signal	f	Wavelength λ	Lumped if smaller than
Lebanese power grid	50 Hz	6000 km	600 km
FM radio	100 MHz	3 m	30 cm
1 GHz radio link	1 GHz	30 cm	3 cm

Three assumptions follow: effects are instantaneous, net charge on each component is zero, no magnetic coupling between components.

SI UNITS

Six units carry this whole course

Quantity	Unit	Symbol	Equals
Charge	coulomb	C	$A \cdot s$
Current	ampere	A	C/s
Voltage	volt	V	J/C
Power	watt	W	$J/s = V \cdot A$
Energy	joule	J	$N \cdot m = W \cdot s$
Resistance	ohm	Ω	V/A

Units cost real missions: NASA lost the Mars Climate Orbiter in 1999 when pound-force seconds met software that expected newton-seconds.

PREFIXES

Pick the prefix that puts the number between 1 and 1000

p

10^{-12}

n

10^{-9}

μ

10^{-6}

m

10^{-3}

k

10^3

M

10^6

G

10^9

0.00001 s → **10 μs**, not 0.01 ms or 10,000 ns.

Your turn: 2,200,000 W = ? 0.000 330 A = ? 0.000 000 047 s = ?

CURRENT

Current is the rate of charge flow

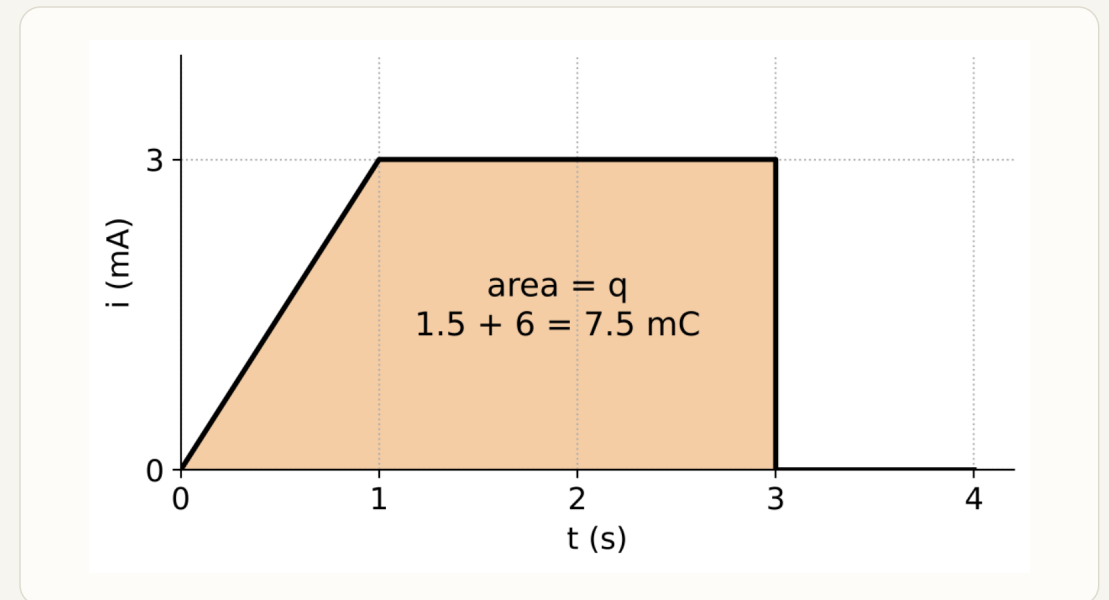
$$i = \frac{dq}{dt}$$

$$q(t) = \int_0^t i(x) dx$$

1 A = 1 C/s. Current is the **slope** of $q(t)$; charge is the **area** under $i(t)$, taking $q = 0$ at $t = 0$.

Graph: $\frac{1}{2}(1 \text{ s})(3 \text{ mA}) + (2 \text{ s})(3 \text{ mA}) = 1.5 + 6 = 7.5 \text{ mC}$.

Nilsson Example 1.2: 5 A for 10 s delivers 50 C.



VOLTAGE

Voltage is energy per unit charge

$$v = \frac{dw}{dq} \qquad 1 \text{ V} = 1 \frac{\text{J}}{\text{C}}$$

Separation of charge

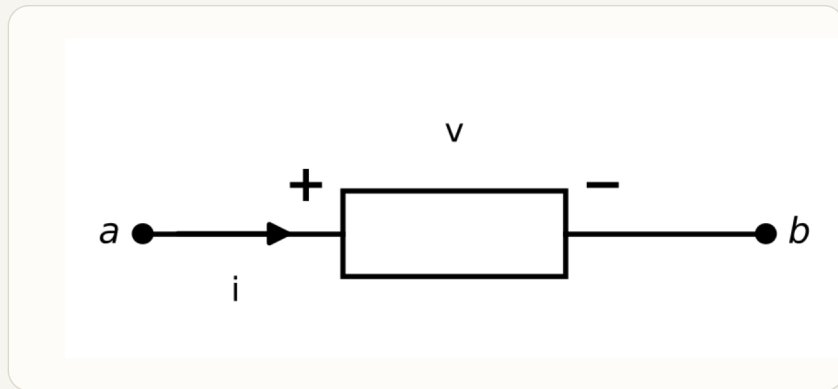
A 12 V battery gives 12 J to every coulomb that passes through it from – to +.

Always between two points

"The voltage at a terminal" means nothing until you say: relative to which other point?

REFERENCE DIRECTIONS

The + / – and the arrow are labels; the sign of the number is the physics



Value	Positive	Negative
v	a is higher than b	b is higher than a
i	+ charge moves along the arrow	+ charge moves against the arrow

You choose the labels freely. Once chosen, every equation must agree with them. Electrons move opposite to the positive-charge flow.

BUILDING BLOCKS

An ideal basic circuit element has three attributes

1

It has only two terminals.

2

It is described by its voltage and current.

3

It cannot be subdivided into other elements.

Ideal = a model, not a part you can buy. Real devices are built from combinations of ideal elements.

POWER

Power is the product of voltage and current

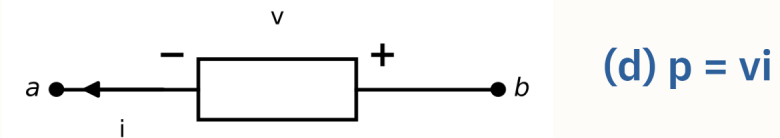
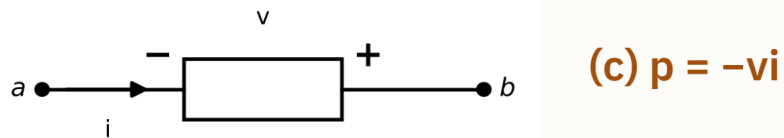
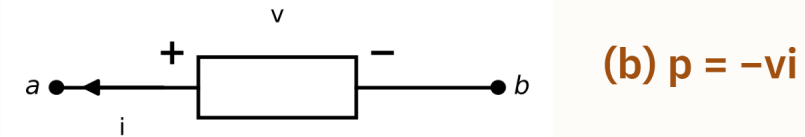
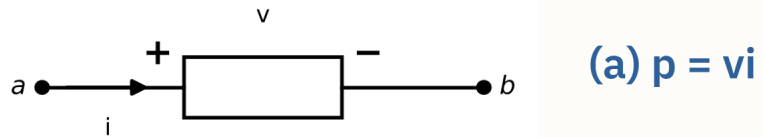
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

1 W = 1 J/s. Power is a property of **one element**: the voltage across it times the current through it.

The useful output is often non-electrical (light, sound, heat, motion), and every real device has a power limit. That is why designs are specified in watts.

THE PASSIVE SIGN CONVENTION

Arrow enters the + terminal: $p = vi$. Otherwise: $p = -vi$



Find the terminal where the current **enters the element**. An arrow pointing away from one terminal means the current enters at the other.

READING THE SIGN

$p > 0$: the element absorbs. $p < 0$: it delivers.

Nilsson Example 1.3: two students analyse the same element with different labels.

Student 1, labels (b)

$$i = 4 \text{ A}, v = -10 \text{ V}$$

$$p = -vi = -(-10)(4) = \mathbf{40 \text{ W, absorbs}}$$

Student 2, labels (c)

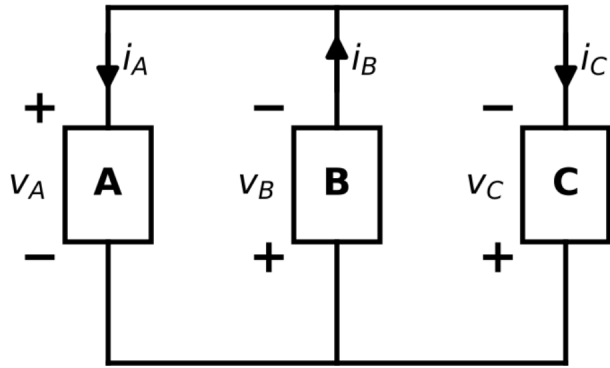
$$i = -4 \text{ A}, v = 10 \text{ V}$$

$$p = -vi = -(10)(-4) = \mathbf{40 \text{ W, absorbs}}$$

Different labels, different numbers, **the same physics.**

PREDICT FIRST · 3 MINUTES, NO CALCULATOR

Which elements absorb power?



Element	A	B	C
v (V)	10	-10	-10
i (A)	2	5	3

1. Does each arrow enter + or -?
2. Absorbs or delivers?
3. Do the three powers add to zero?

ANSWER

A and C absorb; B delivers 50 W

Element	Arrow enters	Power	Result
A	+ (top)	$(10)(2) = 20 \text{ W}$	absorbs
B	+ (bottom): the arrow points up, out of the top	$(-10)(5) = -50 \text{ W}$	delivers
C	- (top)	$-(-10)(3) = 30 \text{ W}$	absorbs

Check: $20 - 50 + 30 = 0$. Delivered 50 W = absorbed 50 W.

ENERGY

Energy is power accumulated over time

$$w(t) = \int_0^t p(x) dx \quad \text{constant } P: \quad w = P t$$

The kWh on your bill

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \text{ MJ}$$

A 2000 W kettle for 3 min

$$2000 \text{ W} \times 180 \text{ s} = 360 \text{ kJ} = 0.1 \text{ kWh}$$

Time in **seconds** gives joules. Hours or minutes are the commonest unit slip.

Power in an element whose voltage and current both decay

Given. $v = 0$ and $i = 0$ for $t < 0$. For $t \geq 0$, with the current entering the + terminal (so $p = +vi$):

$$v = 10 e^{-5000t} \text{ kV} = 10\,000 e^{-5000t} \text{ V}, \quad i = 20 e^{-5000t} \text{ A}$$

Step 1 • multiply

$$p = vi = (10\,000 e^{-5000t}) (20 e^{-5000t}) = 200\,000 e^{-10\,000t} \text{ W}$$

$$e^a e^b = e^{a+b}: \quad -5000t - 5000t = -10\,000t$$

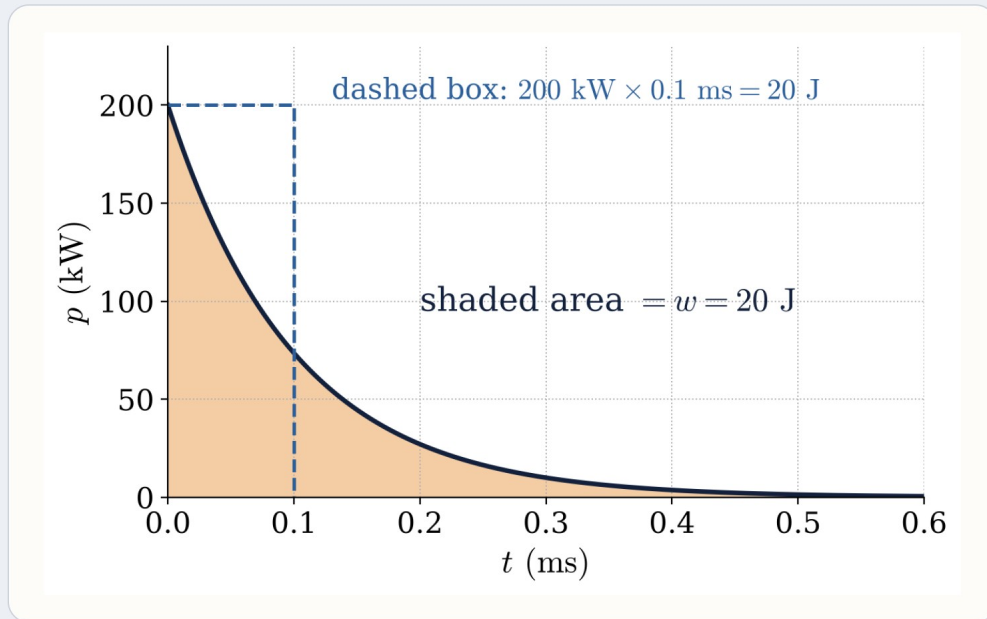
Step 2 • at 1 ms

$$t = 1 \text{ ms}: \quad -10\,000 \times 0.001 = -10$$

$$p(1 \text{ ms}) = 200\,000 e^{-10} = 200\,000 \times 4.54 \times 10^{-5} = 9.08 \text{ W}$$

Convert kV to V before multiplying: $10 \text{ kV} \times 20 \text{ A}$ is $10,000 \text{ V} \times 20 \text{ A} = 200,000 \text{ W}$ at $t = 0$.

The energy delivered is the area under $p(t)$



$$\begin{aligned}
 w &= \int_0^{\infty} p \, dt = \int_0^{\infty} 200\,000 e^{-10\,000t} \, dt \\
 &= \left[-\frac{200\,000}{10\,000} e^{-10\,000t} \right]_0^{\infty} = \left[-20 e^{-10\,000t} \right]_0^{\infty} \\
 &= 0 - (-20) = 20 \text{ J}
 \end{aligned}$$

Check. The area under a decaying exponential equals its starting value times its time constant, here $1/10,000 \text{ s} = 0.1 \text{ ms}$:

$$p(0) \times \frac{1}{10\,000} \text{ s} = (200\,000 \text{ W})(0.1 \text{ ms}) = 20 \text{ J}$$

Keeping $-5000t$ in the exponent of p gives 40 J, twice the correct value.

YOUR SELF-CHECK

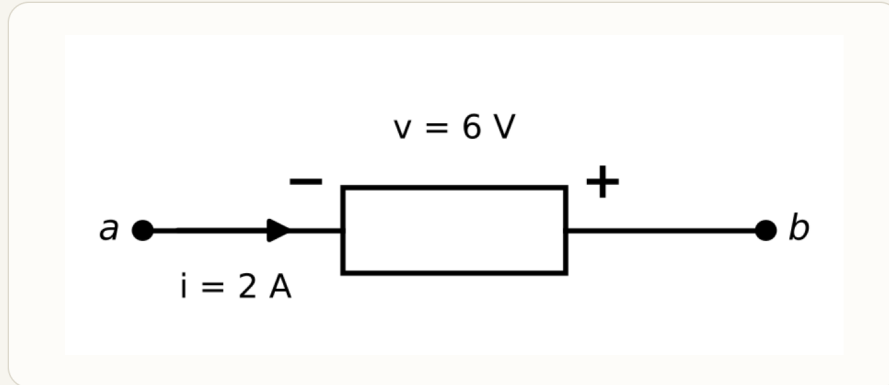
$$\sum p = 0$$

In every circuit, the power delivered equals the power absorbed. If your powers don't add to zero, a sign is wrong somewhere.

Every booklet problem in Chapter 2 ends with a power balance. Nilsson's Practical Perspective uses it to check a model of a home's supply.

CHECKING AN AI ANSWER

" $p = (6)(2) = 12 \text{ W}$, absorbed." What is wrong?



In a 2025 study, misread polarities and current directions caused about a third of Gemini 2.5 Pro's errors on undergraduate circuit problems. That is today's skill. Use AI to check your work, then check its signs.

The arrow enters the – terminal, so $p = -(6)(2) = -\mathbf{12 \text{ W}}$. The element **delivers** 12 W.

WHAT COMES NEXT

Practise openly, then show it on your own

Now, in class

Booklet Problems 1.1 and 1.2.

At home

Booklet 1.3 and the 16-item practice sheet. AI tools allowed; not marked.

Next session

15-minute quiz, no devices. Every question is a twin of a practice-sheet question.

If you can do the sheet on your own, you will do well on the quiz.

CHECKLIST

You can do these without help

- Write any value in engineering notation with the right prefix.
- Find charge from a current graph (area), and current from a charge function (slope).
- Say what a negative v or i means physically.
- Write $p = v i$ or $p = -v i$ from a diagram, and say absorbs or delivers.
- Compute energy for constant and for time-varying power.
- Check an answer with a power balance.